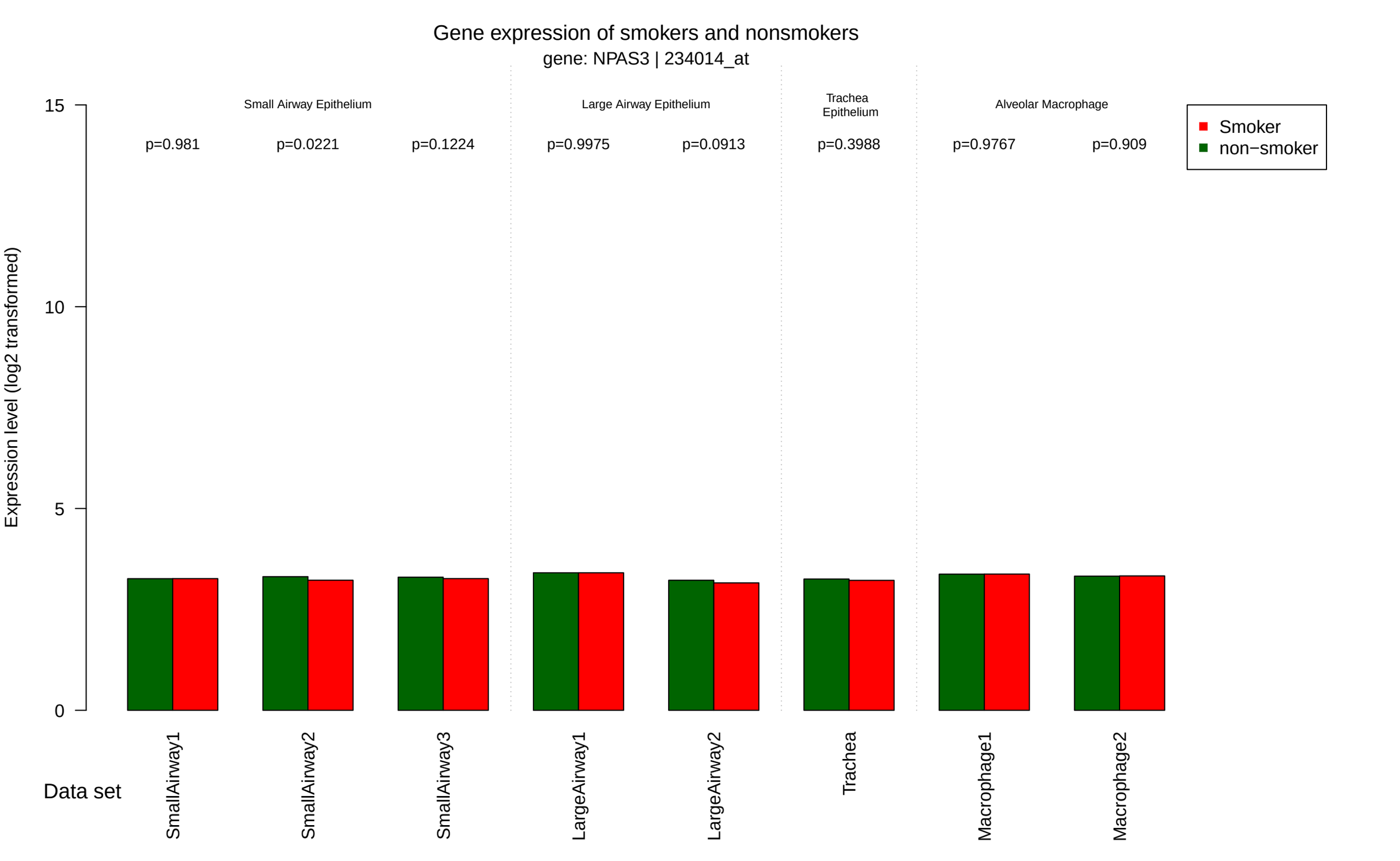
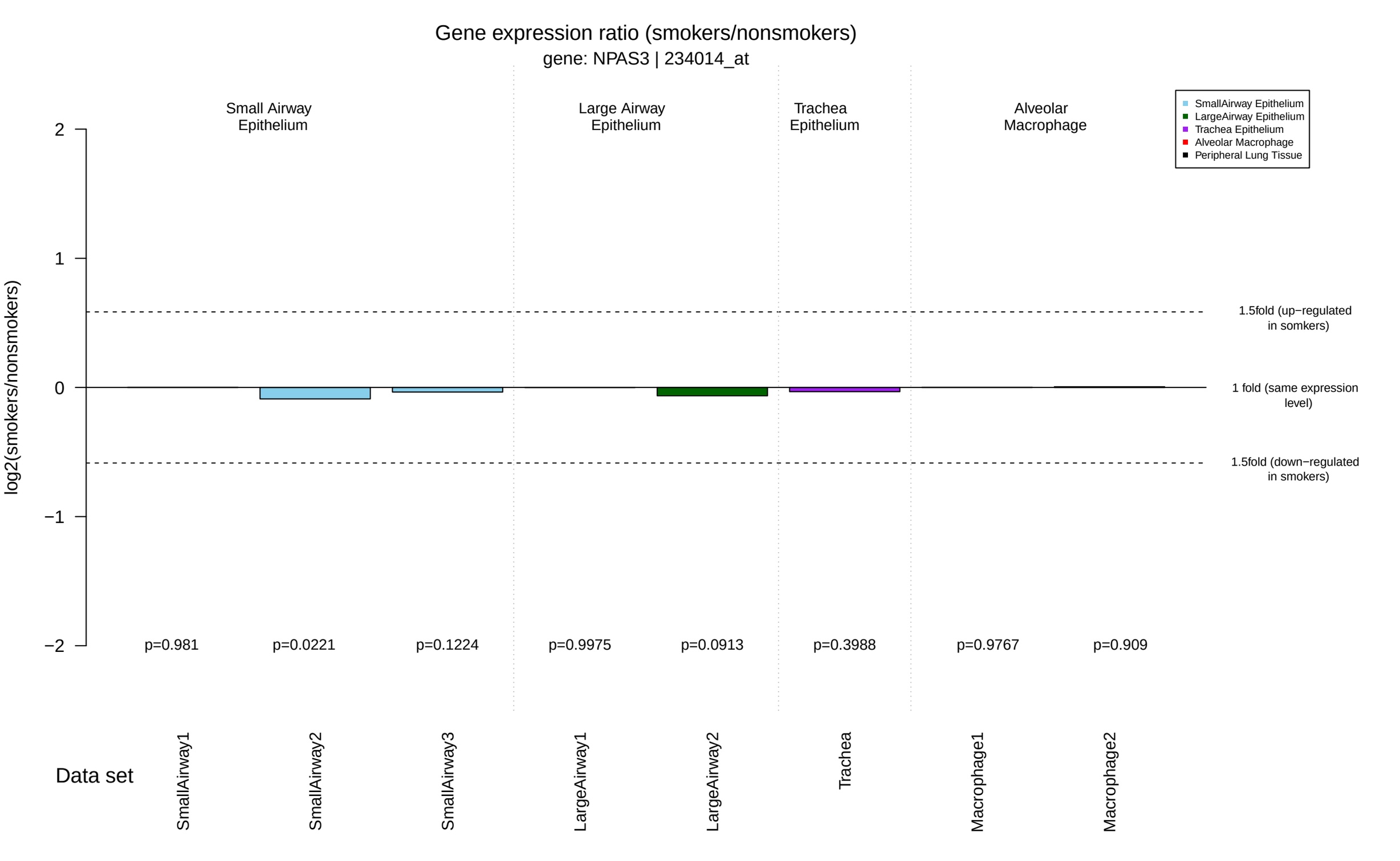
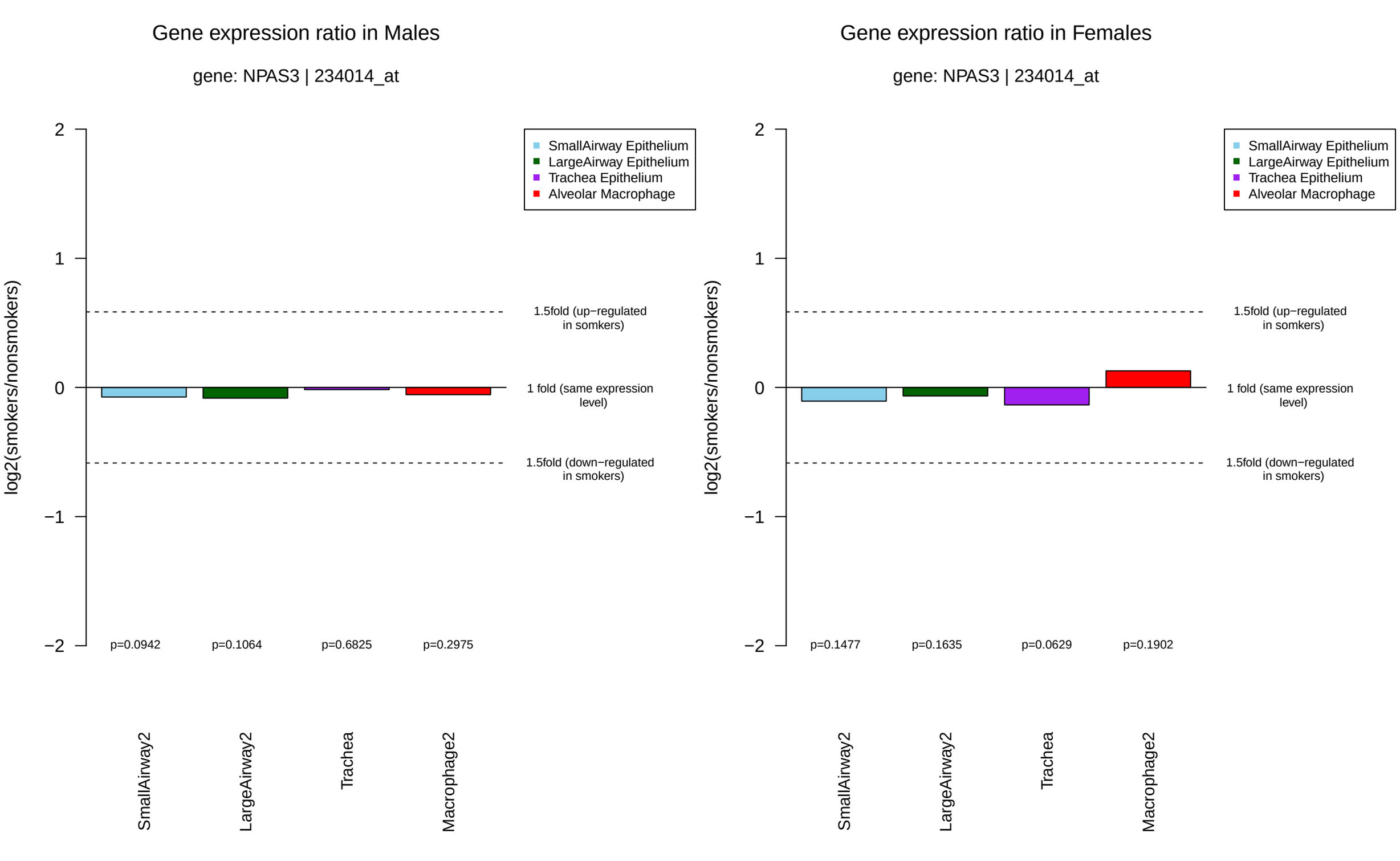
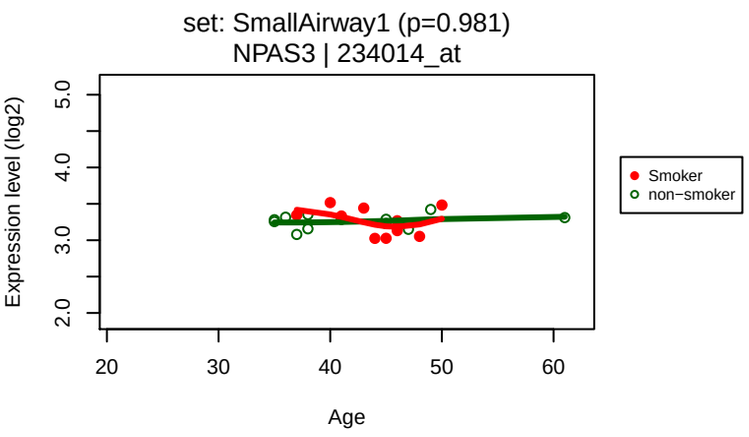
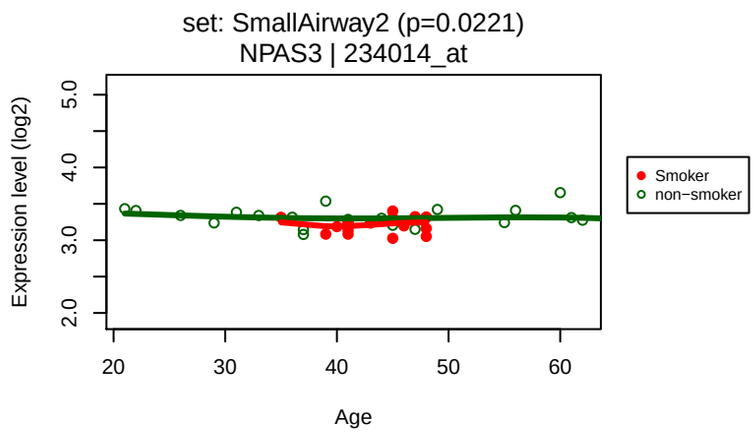
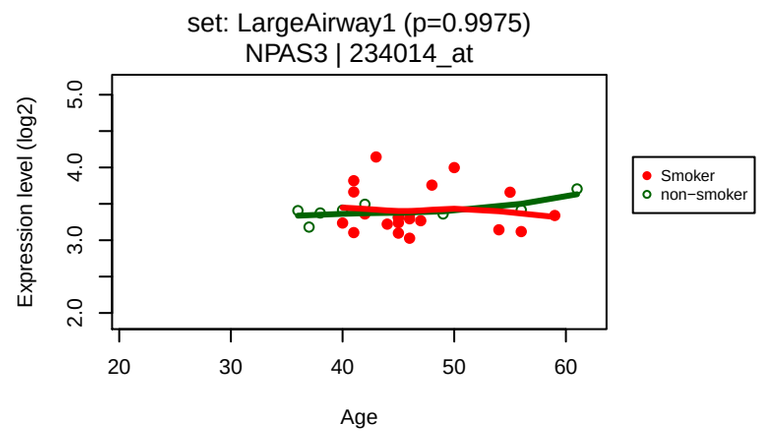

Gene expression of smokers and nonsmokers
gene: NPAS3 | 234014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.981
p=0.0221
p=0.1224
p=0.9975
p=0.0913
p=0.3988
p=0.9767
p=0.909
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

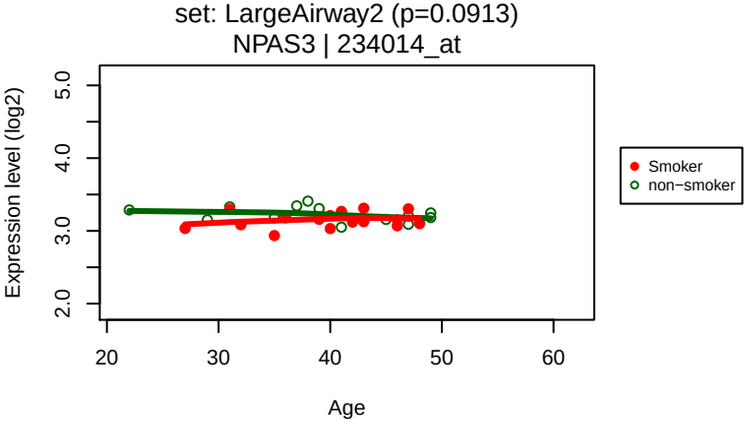

Gene expression ratio (smokers/nonsmokers)
gene: NPAS3 | 234014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.981
p=0.0221
p=0.1224
p=0.9975
p=0.0913
p=0.3988
p=0.9767
p=0.909
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

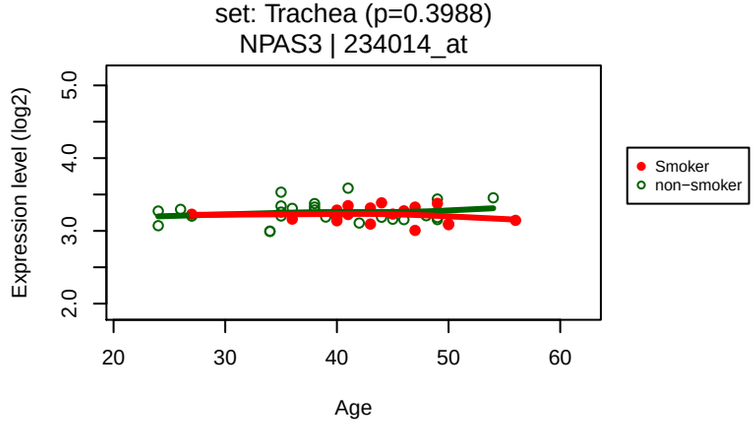

Gene expression ratio in Males
gene: NPAS3 | 234014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0942
p=0.1064
p=0.6825
p=0.2975
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAS3 | 234014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1477
p=0.1635
p=0.0629
p=0.1902
SmallAirway2
LargeAirway2
Trachea
Macrophage2

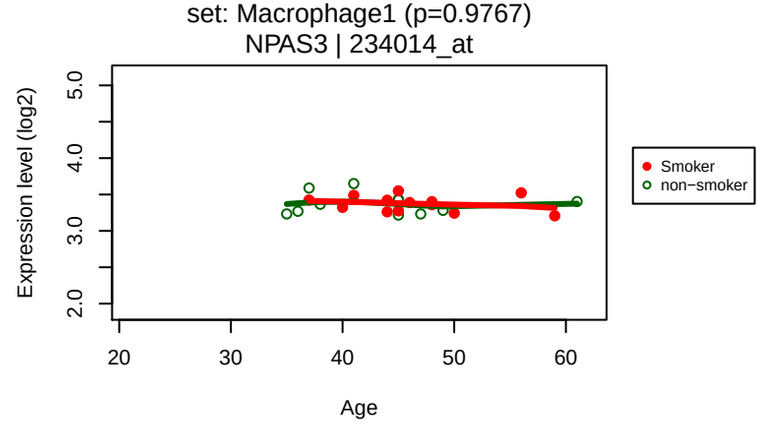

set: SmallAirway1 (p=0.981)
NPAS3 | 234014_at
Expression level (log2)
Age
Smoker
non−smoker

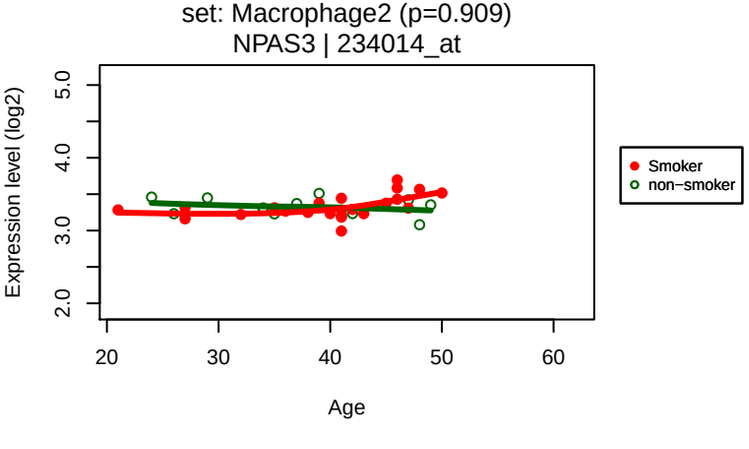

set: SmallAirway2 (p=0.0221)
NPAS3 | 234014_at
Expression level (log2)
Age
Smoker
non−smoker

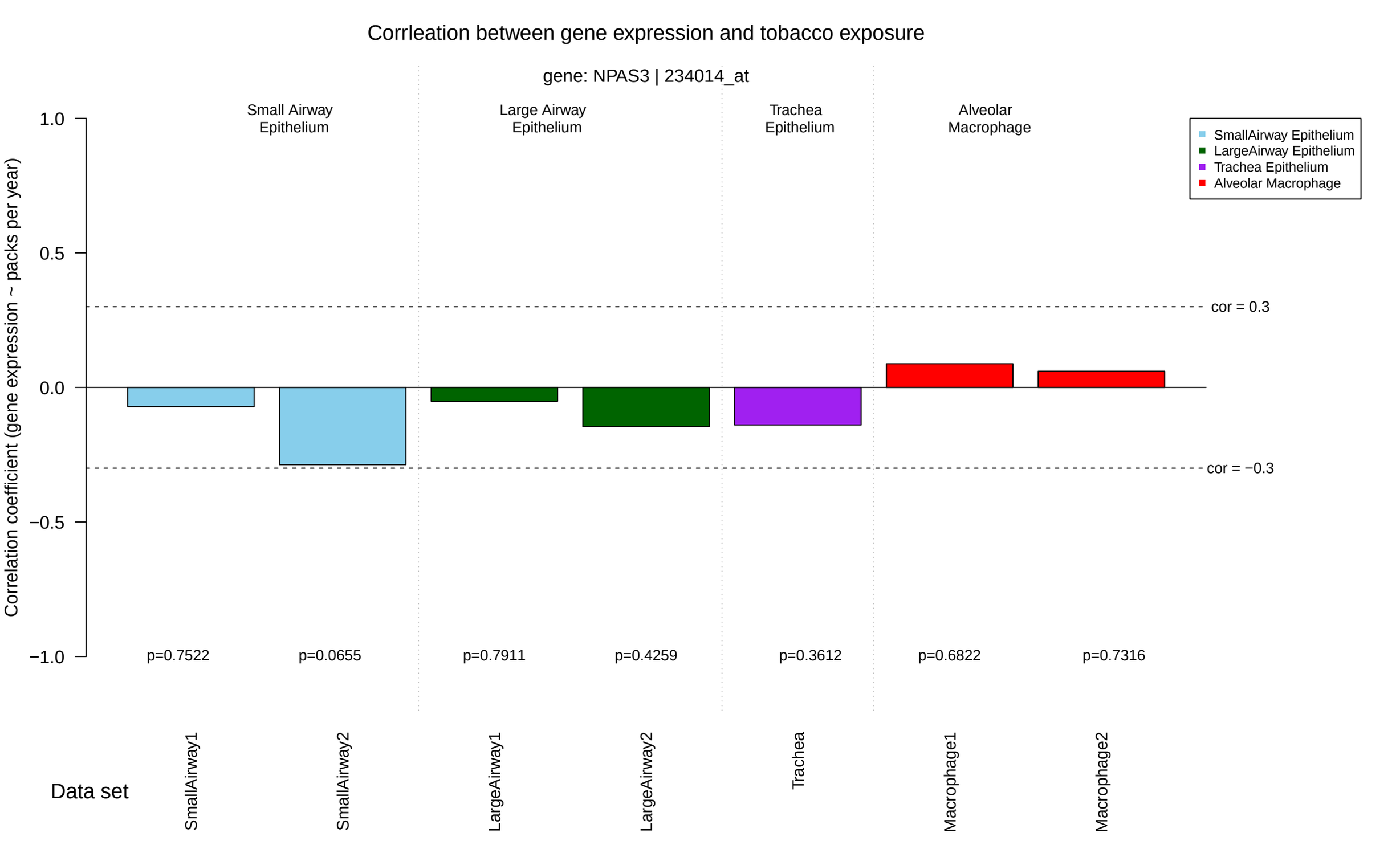

set: LargeAirway1 (p=0.9975)
NPAS3 | 234014_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0913)
NPAS3 | 234014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3988)
NPAS3 | 234014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9767)
NPAS3 | 234014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.909)
NPAS3 | 234014_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAS3 | 234014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7522
p=0.0655
p=0.7911
p=0.4259
p=0.3612
p=0.6822
p=0.7316
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2